

OPENING OF COAL SEAMS

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A solid orange horizontal bar at the bottom of the slide.

Underground Mining Methods

*Original
land surface*

Dragline removing
mountain top

Surface Mining Methods

rock spoil

Dozer along
contour bench

coal beds

Auger mining

Dragline in
pit

Drift mine

rock spoil

Slope mine

coal beds

Shaft mine

S.G.

MINING METHODS AND MINING ENTRIES

OPENCAST VS UNDERGROUND METHODS

ADVANTAGES

- HIGHER PRODUCTIVITY
- GREATER OUTPUT PER MINE
- GREATER CONCENTRATION
- LOWER CAPITAL COST
- LOWER OPERATING COST
- SHORTER GESTATION PERIOD
- POSSIBILITY OF MINING LOWER GRADE ECONOMICALLY
- GREATER RECOVERY / INCREASED MINEABLE RESERVE
- LESS LIMITATION OF SIZE & WEIGHT OF MACHINES
- GREATER SAFETY
- BETTER WORK ENVIRONMENT
- LESS INFLUENCE OF GEOLOGICAL UNCERTAINTY
- SIMPLIFIED ENGINEERING, PLANNING AND CONTROL

DISADVANTAGES

- ADVERSE EFFECT ON ENVIRONMENT AND ECOLOGY
- PROBLEMS OF NOISE, VIBRATION, FUME, DUST ETC. TO COMMUNITY
- LAND PROFILE GREATLY DAMAGED
- LARGE WATERLOGGED WORKING LEFT
- HIGH RECLAMATION COST
- HANDLING OF LARGE QUANTITY OF WASTE MATERIAL, ENERGY CONSUMPTION
- SOMETIMES HIGH INITIAL INVESTMENT
- ACCUMULATION OF NOXIOUS GASES IN DEEP MINES
- DIFFICULT TO GET ENVIRONMENTAL CLEARANCE
- REQUIRES EVACUATION AND RESETTLEMENT OF HABITANTS
- AFFECT SURFACE FEATURES LIKE ROADS, OH MAINS, RIVERS, RAILWAYS
- ILLEGAL MINING

OPENING UP OF COAL SEAMS

- **LEGAL REQUIREMENTS:**
- **REG 71 of CMR, 2017:**
- **(1) No person shall be employed, or be permitted to enter or remain**
- **for purposes of employment, in any working belowground, unless the working is provided with**
- **at least two shafts, inclines or other outlets to the surface –**
- **(a) with which every seam or section of the time being at work has a communication so as to**
- **afford separate means of ingress and egress to the persons employed therein;**
- **(b) which do not have their surface openings in the same building; and**
- **(c) which are under the sole control of the manager :**
- **Provided that the Chief Inspector may, by an order in writing and subject to such conditions**
- **as he may specify therein, permit the employment belowground of persons even in a case**
- **where the two shafts, inclines or outlets are not under the control of the same manager.**

OPENING UP OF COAL SEAMS

- **LEGAL REQUIREMENTS:**
- **REG 71 of CMR, 1957:**
- **(6) Such shafts, inclines or outlets shall not be less than 13.5 metres distant from one another**
- **at any point, and each shall be connected with the other by means of a walkable passage, not less than 1.8 metres high 1.5 metres wide, through the workings belowground that are being served by such shafts, inclines or outlets.**

TYPES OF ENTRIES INTO MINES

- VERTICAL SHAFTS
- INCLINE SHAFT / SURFACE DRIFTS
- ADITS

VERTICAL SHAFTS vs SURFACE DRIFTS / INCLINES

PARAMETERS	SHAFT	INCLINES
TIME REQUIREMENT FOR COMPLETION	MORE	LESS
CAPITAL COST	MORE	LESS
CONTINUOUS TRANSPORT OF MINERAL	CYCLIC	CONTINUOUS
CAPACITY OF COAL TRANSPORT	LIMITED	LARGE
OPERATIONAL COST	MORE	LESS
RESISTANCE TO VENTILATION	LESS	MORE
LENGTH OF CABLE, PIPELINE	LESS	MORE
POSSIBILITY OF IMPACTED BY GEOLOGICAL UNCERTAINTY	LESS	MORE
LOSS OF COAL IN PROTECTIVE PILLAR	LESS	MORE
 ENVIRONMENTAL PROBLEM	MORE	LESS

ACCESS BY ADITS

- HILLY TERRAIN
- LEVEL DRIFT FROM FOOT OF HILL TO INTERSECT THE SEAM
- OUTCROP ABOVE LOADING LEVEL
- SUFFICIENT SPACE AVAILABLE AT MOUTH OF ADIT TO FACILITATE CONSTRUCTION OF SURFACE INFRASTRUCTURE
- CHEAPEST MODE OF ENTRY
- CAN BE EQUIPPED WITH LOCOMOTIVE HAULAGE , CONVEYOR

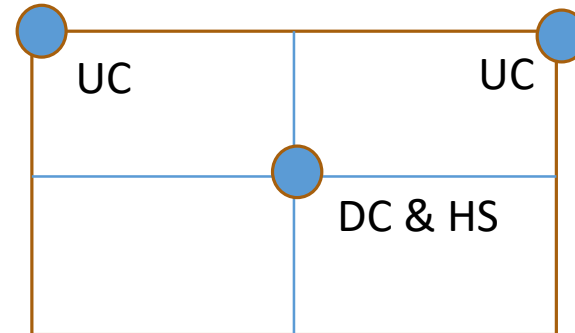
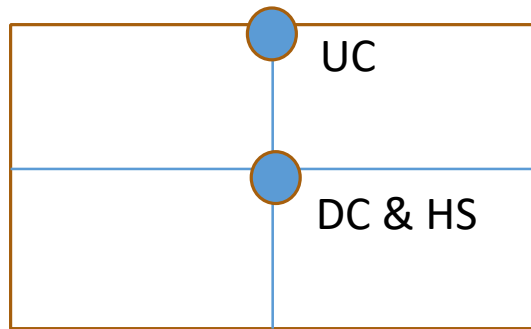
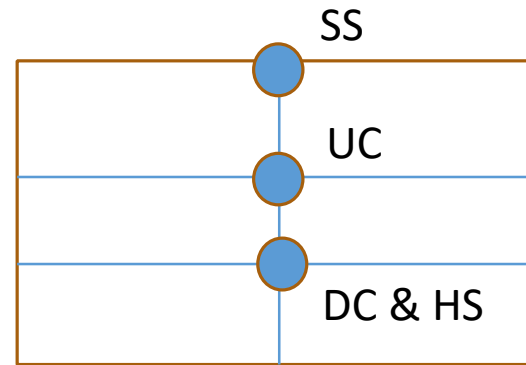
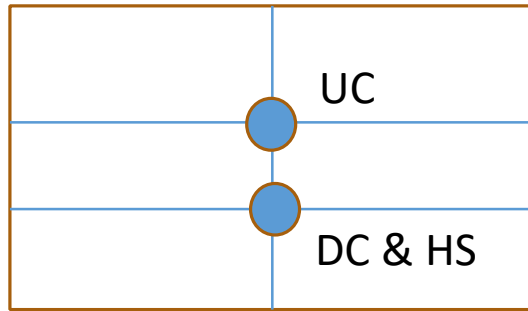
LOCATION OF INCLINES

- SURFACE FACTORS:
- Level of entry: The point at which a body of rising water on surface can enter the mine should be above HFL by atleast 1.5m.
- Reg 149. Danger from surface water –
- (1) (a) Where any mine or part thereof is so situated that there is any danger of inrush of surface water into the mine or part, adequate protection against such inrush shall be provided and maintained. Whether such protection is adequate or not can be determined by the Chief Inspector or whose decision shall be final.
- (b) Except with the permission of the Chief Inspector in writing and subject to such conditions as he may specify therein and subject to the provisions of clause (a), every entrance into a mine shall be so designed, constructed, and maintained that its lowest point (which means the point at which a body of rising water on surface can enter the mine) shall be not less than 1.5 metres above the highest flood level at that point

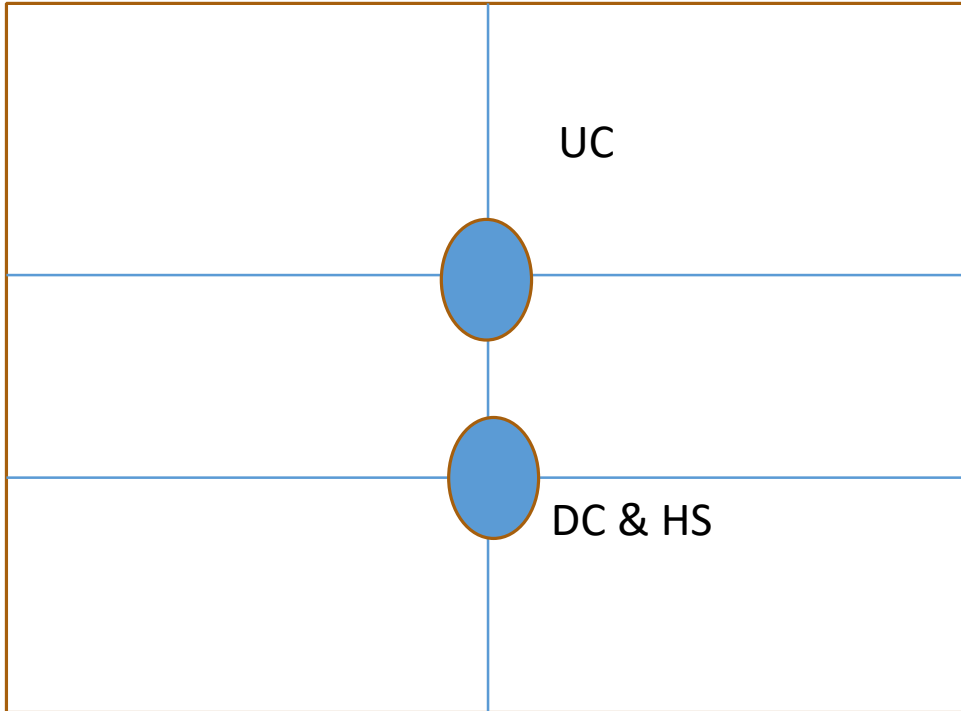
LOCATION OF INCLINES

- SURFACE FACTORS:
 - Level ground, well drained, requiring little or no cutting or filling
 - Sufficient area for construction of surface infrastructure
- GEOLOGICAL FACTORS:
 - Driving through geologically disturbed area should be avoided
- MINING FACTORS:
 - Preferably through middle of the property to divide in two equal halves
- OPERATIONAL FACTORS: Nearer to
 - Road and rail access
 - Public utility services
 - Workshop, coal washing plant

Various locations of vertical shafts



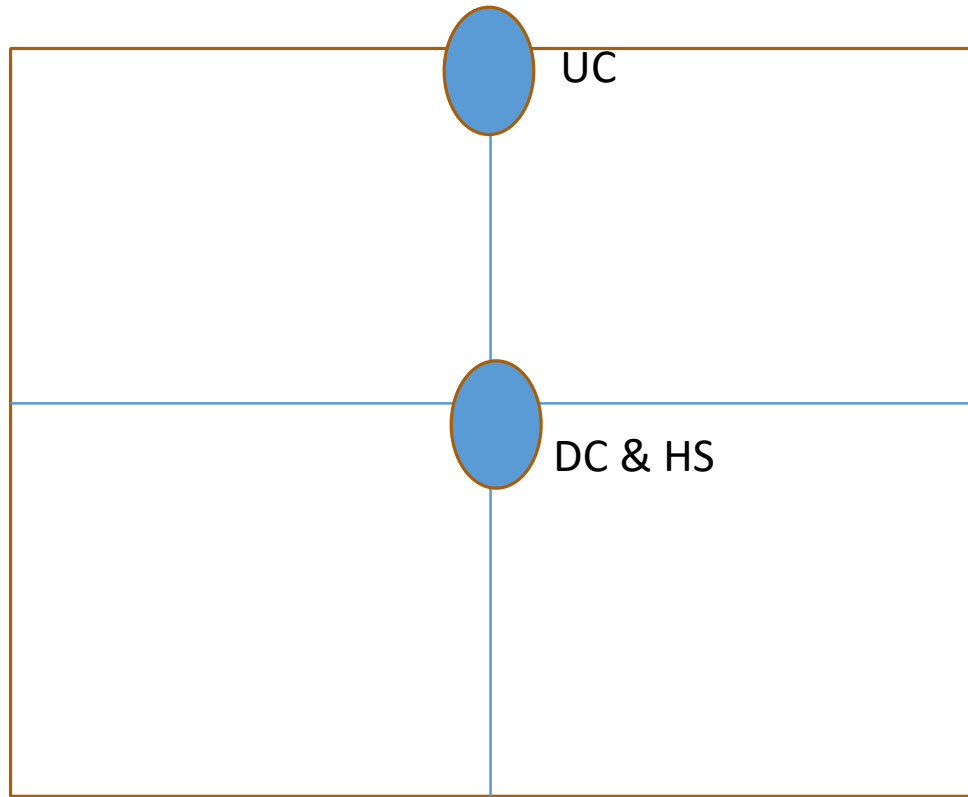
Various locations of vertical shafts



Two shafts at the centre

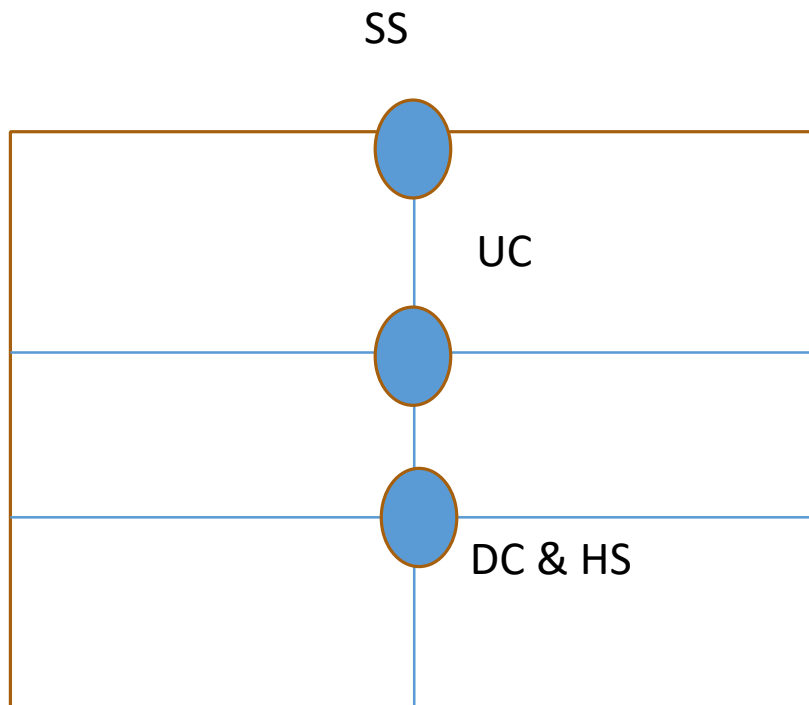
- Easy to connect two shafts UG
- Compact surface layout
- Less area of shaft pillar left
- In case of emergency, winding of coal through man winding shaft or vice-versa

Various locations of vertical shafts



Ventilation shaft shallower
Ascentional ventilation throughout life
of the mine

Various locations of vertical shafts

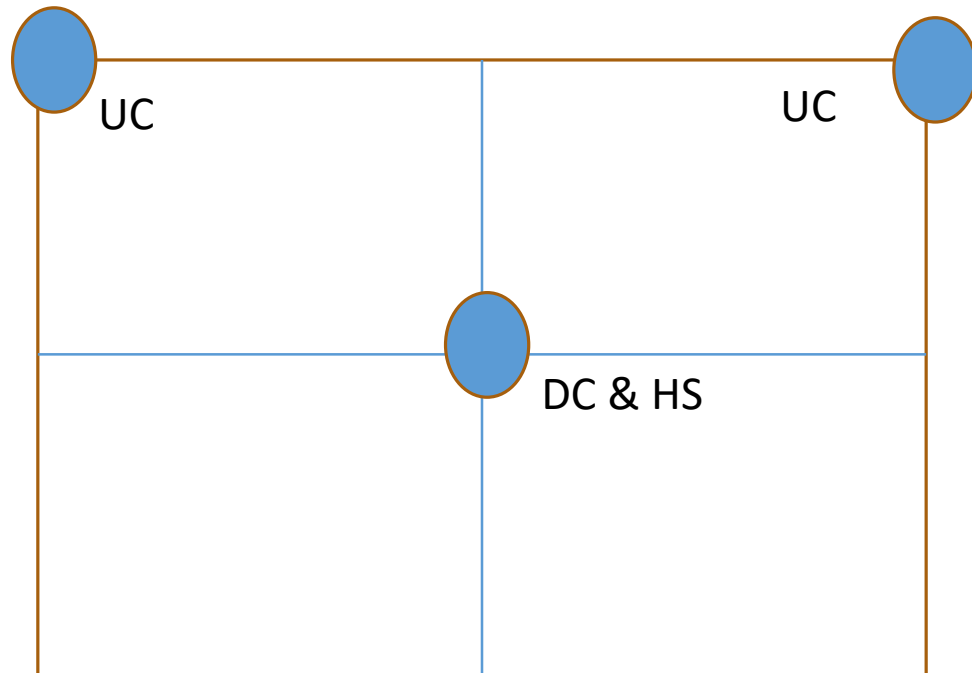


Same as Model 1

Third shaft on rise most boundary for
transport of slurry

Additional shaft to be sunk

Various locations of vertical shafts



Equal length of ventilation circuit
Surface arrangements spread